

THE POWER SHELF EATS THE COMPUTE: WHY THE AI RACK MUST MOVE TO 800 VDC

Technical write-up | Data centre infrastructure | Independent research
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The trend	Rack power is scaling faster than the in-rack architecture that distributes it. The 54 VDC busbar design, standard since Hopper, begins hitting physical limits above 200 kW per rack. NVIDIA Kyber, due 2027, is specified at 576 GPUs and up to 1 MW.
The claim	At megawatt scale the incumbent architecture consumes the rack it is meant to power. The constraint has moved from the accelerator into the last fifty feet of power delivery, and that is a power electronics problem before it is a compute problem.

The arithmetic that kills the 54 volt rack

Today's AI racks distribute power internally at 54 VDC, the design NVIDIA has used since Hopper and early Blackwell, with copper busbars carrying current from rack-mounted power shelves to the compute trays. It works at tens of kilowatts and it fails on physics above roughly 200 kW, which is the threshold NVIDIA itself names. Two numbers make the failure concrete. A single 1 MW rack distributed at 54 VDC requires up to 200 kg of copper busbar, and the rack busbars alone inside a 1 GW campus would run to roughly 200,000 kg of copper. The harder limit is spatial rather than material. A GB200 or GB300 NVL72 already carries up to eight power shelves, and holding 54 VDC into a megawatt-scale Kyber rack would demand up to 64U of power shelves. NVIDIA's own framing is that this leaves no room for compute. That is the entire argument in one line: the power delivery architecture starts eating the thing it exists to power.

800 volts is an electric vehicle supply chain arriving in the data centre

The fix is to convert fewer times and carry the power at higher voltage. The published architecture rectifies 13.8 kV AC grid power directly to 800 VDC at the facility perimeter, then feeds two-conductor 800 V into the rack and does DC/DC conversion at the compute tray. Moving row distribution from 415 VAC to 800 VDC pushes 85% more power through the same conductor size and cuts backbone copper by about 45%. End-to-end efficiency improves by up to 5% against 54 V systems, maintenance cost falls by up to 70% because there are far fewer power supply units and fans left to fail, and NVIDIA puts total cost of ownership up to 30% lower. The reason this is happening now rather than a decade ago has little to do with data centres. The component base for high-voltage DC switching was built and qualified at volume by electric vehicle charging, and the data centre is inheriting it.

This is the interconnection argument observed one layer down

I have argued elsewhere that the binding constraint on AI buildout moved from generation capacity to grid connection. This is the same movement at rack scale. Each accelerator generation converts more of the problem out of silicon and into power delivery and heat rejection, so what gates a Kyber deployment in 2027 is the rectifier, the busway, the DC/DC stage and the heat path rather than the chip roadmap. The two constraints are also coupled: a campus that cannot energise its connection never installs a 1 MW rack, and a campus holding an energised connection but no power electronics cannot use it. The distinction worth holding is duration. Silicon shortages clear in quarters because fabs and lines can be added. Power delivery clears in years because it is gated by transformers, switchgear and the grid interface. They do not resolve together, and pricing them as one cycle is the error.

Where the value accrues if the claim is right

NVIDIA published its partner list for this transition, and a partner list is revealed preference rather than marketing. It splits three ways. At the systems layer, Eaton, Schneider Electric and Vertiv, which is the same set that has been acquiring cooling assets. At the component layer, Delta, LiteOn, Megmeet, Flex Power and Lead Wealth. At silicon, Infineon, Texas Instruments, onsemi, ROHM, STMicroelectronics, Renesas, Analog Devices and MPS, alongside the two wide-bandgap specialists Navitas and Innoscience. The honest read is that 800 VDC is a wide-bandgap semiconductor story at the component layer and a power systems story at the rack layer, and the second is far easier to underwrite than the first. Two second-order effects matter more than they appear. NVIDIA states it is still weighing traditional transformers against solid-state transformers for the facility interface, which is a genuinely open architectural fork. And energy storage is written into the 800 VDC specification itself, sized for load spikes and subsecond GPU power swings rather than for arbitrage, which makes batteries rack-adjacent infrastructure rather than only a grid-side trade. On copper the nuance runs against the headline: 45% less copper per unit of power does not mean less copper, because power is growing faster than the saving. It changes the specification and the mix, not the direction.

What would prove me wrong, and what I am watching

The strongest counterargument is that a published standard is not a deployment. NVIDIA and roughly thirty power electronics vendors have coordinated this through the Open Compute Project, but the party that has to rebuild its electrical room is the colocation operator, and operators have said very little publicly about adopting it. NVIDIA names the alternative itself: a dedicated rack of power supplies beside every compute rack. If the industry takes that path, the transition slows, stays partial, and most of this decays into a component refresh rather than an architectural change. I would also be wrong if rack density plateaus below the 200 kW line because inference economics favour many smaller deployments over few enormous ones, which removes the forcing function entirely. And 2027 is a date in an industry where dates slip. So I watch four things: public 800 VDC commitments from named colocation operators rather than from vendors, whether solid-state transformers take the facility interface, wide-bandgap capacity announcements aimed at data centre rather than automotive demand, and whether GB300-class racks in the field are landing above 200 kW or comfortably below it.

Sources: NVIDIA Technical Blog, "NVIDIA 800 VDC Architecture Will Power the Next Generation of AI Factories", 20 May 2025; NVIDIA 800 VDC architecture pages; Open Compute Project. All quantitative figures are as published by NVIDIA on those dates. Draws on my own notes on data centre interconnection (15 August 2026), grid scale battery storage (7 August 2026) and copper (29 July 2026). Personal research, not investment advice.